Evaluating Evolutionary and Gradient- Based Algorithms for Optimal Pathfinding

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Introduction

While numerous algorithms offer potential solutions, their comparative efficiency and reliability when confronted with nonlinear terrains remain to be systematically evaluated. This study assesses three pathfinding algorithms—Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Sequential Quadratic Programming (SQP)—to establish a basis for comparison in terms of efficiency and computational speed.

Results from twenty simulations indicate that SQP achieves lower path costs and reduced computational time than GA and PSO. In particular, SQP demonstrates reduced variability in path costs and quicker convergence to optimal paths, proving more effective in nonlinear environments. In contrast, GA and PSO showed higher variability in their path cost distributions.These results suggest gradient-based SQP as a preferable solution for complex pathfinding tasks.

Methods

The domain for path traversal was constrained within [−3, 3] arbitrary units on both x and y axes, with the appropriate cost function value along the z-axis. A fixed number of ten waypoints was chosen to balance path accuracy and optimization time. The cost function for traversing the environment was evaluated by interpolating segmenst between consecutive waypoints into smaller segments of 0.1, summated, and multiplied by the distance to the subsequent interpolated point.

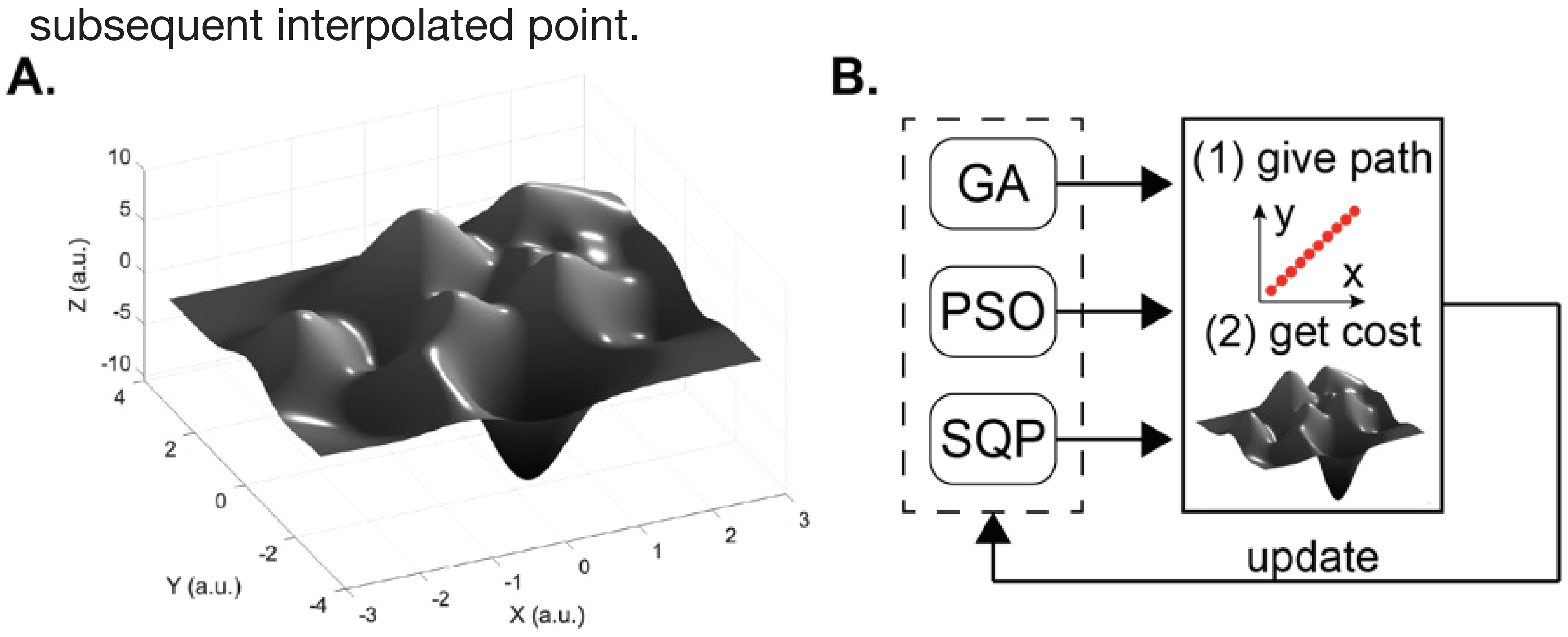

Figure 1. The environment to traverse (A) and the simulation pipeline (B). The simulation procedure entailed refining the path predictions by each algorithm separately using the landscape-dependent cost function. Abbreviations: GA—genetic Algorithm; PSO—Particle Swarm Optimization; SQP—Sequential Quadratic Programming; a.u.—arbitrary units.

The first algorithm applied was the Genetic Algorithm (GA), a search method that mimics the process of natural selection. The algorithm was initiated with a randomly generated population of solutions (set to 100), each comprising ten waypoints that defined the path. The algorithm was configured to execute over a maximum of 100 generations, during which selection, crossover (crossover fraction was set to 0.8), and mutation (random mutation percentage was chosen from a Gaussian distribution) were applied to each generation to refine the population towards the optimal path.

Second algorithm, Particle Swarm Optimization (PSO), relied on collective intelligence of "particles", simultaneously exploring the environment to find minima. The movement of particles through the environment varies by their position and velocity in the environment. To ensure parity in computational effort with the GA, we configured the PSO with a swarm of 100 particles and limited the optimization process to the same number of iterations.

The third algorithm used was the Sequential Quadratic Programming (SQP) procedure ("fmincon" function in the MATLAB's Optimization Toolbox), which operates by iteratively solving approximations of the function that combines the cost and constraints. Here, the algorithm was initiated with an initial guess defined as a straight path trajectory between start and end points. To ensure fair comparison across the algorithms, the total number of function evaluations was capped at 9605, matching the evaluation number used by GA at 100 generations.

Results

The performance of each algorithm was assessed by evaluating (1) precision, measured by the cost of the computed path, and (2) simulation execution time, assessed over 20 simulation sessions. Paths computed by each algorithm are shown in Figure 2.

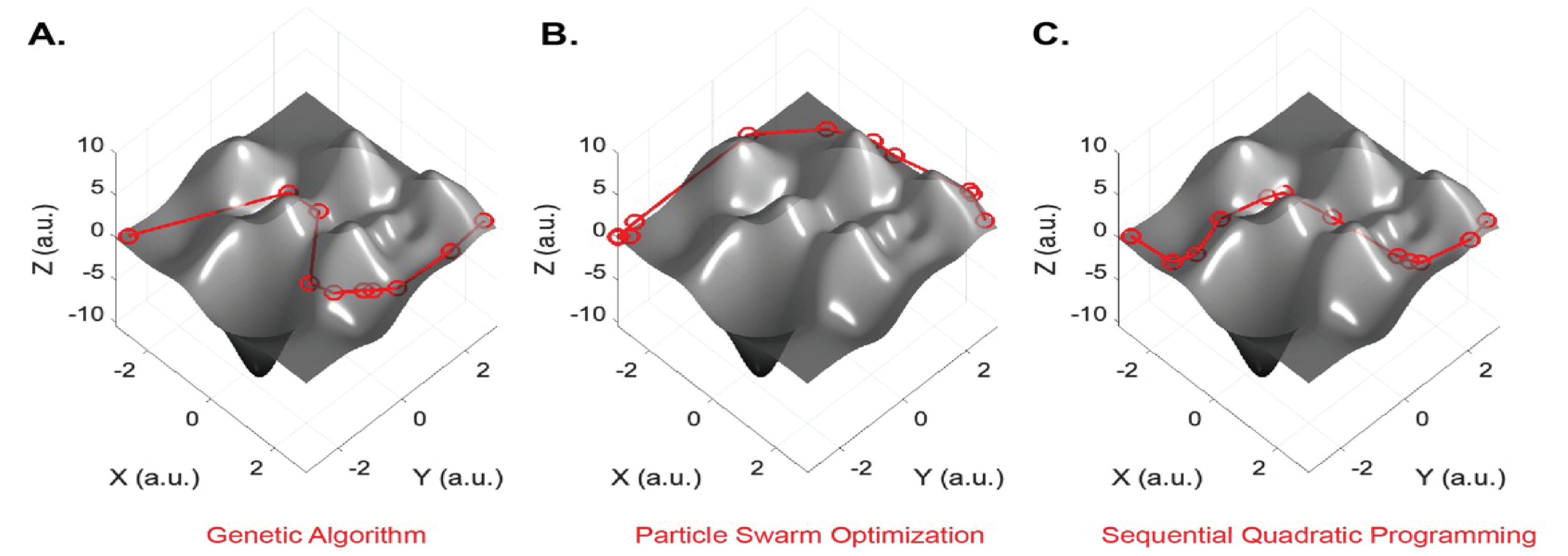

Figure 2. Representative paths calculated by the three evaluated algorithms: (A) Genetic Algorithm (GA), (B) Particle Swarm Optimization (PSO), and (C) Sequential Quadratic Programming (SQP), each illustrating the distinct pathfinding solutions. Abbreviations are the same as in Figure 1.

Figure 3A presents the comparisons between the final costs of the paths computed in each of the 20 simulations conducted by the three methods.

The performance of the GA showed the greatest variability, with a standard deviation (STD) of 1.9381 a.u.; for PSO and SQP, the STDs were 0.9035 a.u. and 0.1199 a.u., respectively.

SQP procedure demonstrated the significantly higher performance in both high accuracy (median of 17.46 a.u.) and simulation execution time (median: 0.8309 min per cycle session); for GA and PSO the performance in accuracy and simulation execution time medians were 18.14 a.u.; 11.02 min per cycle session and 18.38 a.u.; 12.48 min per cycle session, respectively.

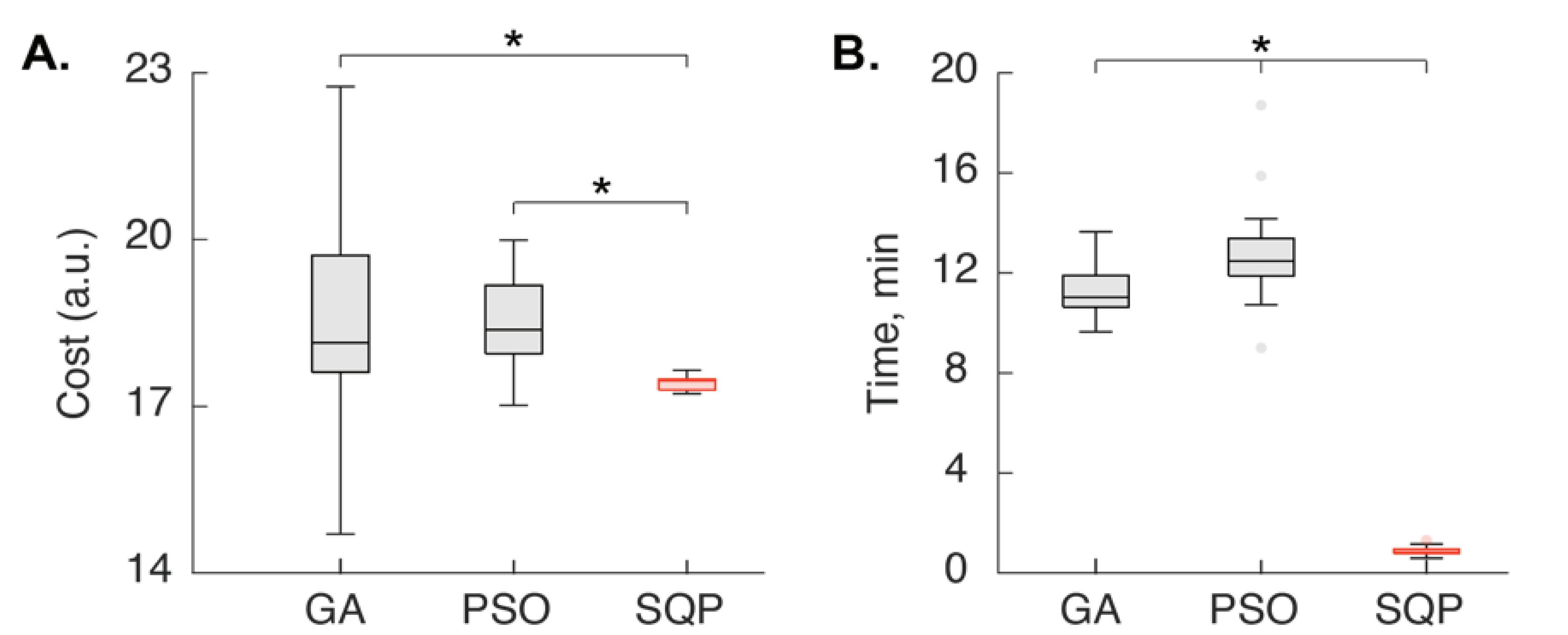

Figure 3. Distributions of path costs (A) and simulation execution times (B) for 20 simulations across three methods. The algorithm with superior performance is highlighted in red in each subplot. Asterisks denote statistically significant differences. Boxes show the interquartile range, bounded by the 25th and 75th percentiles, with whiskers extending to the non-outlier minimum and maximum values. Outliers are indicated as individual points. Abbreviations are the same as in Figure 1.

Discussion

SQP consistently demonstrated superior performance with the lowest variability and fastest execution times, attributed to its gradient-based methodology that ensures quicker convergence. In contrast, GA and PSO showed higher variability in their path cost distributions, with PSO in particular showing the longest execution times due to its complex iterative calculations and potential to get stuck in local optima.

References

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